Density-Functional Theory Exemplified by the Quantum Rabi Model

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The Quantum Rabi Model

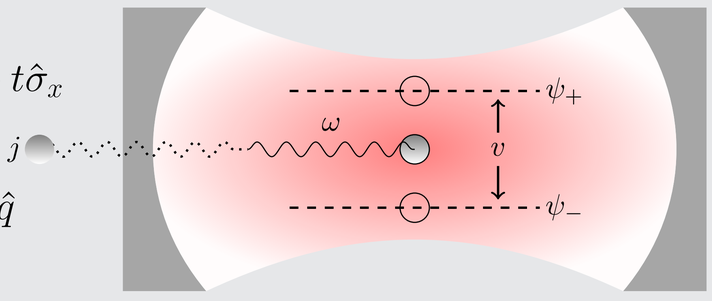
Internal Hamiltonian

$$\hat{H}_0^\lambda = \frac{1}{2}\hat{p}^2 + \frac{\omega^2}{2}\hat{q}^2 + \lambda g \hat{\sigma}_z \hat{q} - t\hat{\sigma}_x$$

Full Hamiltonian

$$\hat{H}^\lambda(v,j) = \hat{H}_0^\lambda + v\hat{\sigma}_z + j\hat{q}$$

"Potentials" $v \in \mathbb{R}$ and $j \in \mathbb{R}$



DFT vs. QRabi DFT

	DFT	QRabi DFT
Density	$\rho(\mathbf{r})$	(σ, ξ)
Potential	$v(\mathbf{r})$	(v, j)
N-rep.	$\mathcal{I}_N$	$[-1,1] \times \mathbb{R}$
v-rep.	unknown	$(-1,1) \times \mathbb{R}$

"Densities" in QRabi:

$\sigma = \langle\psi|\hat{\sigma}_z|\psi\rangle$ (polarisation)

$\xi = \langle\psi|\hat{q}|\psi\rangle$ (displacement)

QRabi DFT formulation

Energy functional

$$E^\lambda(v,j) = \inf_\psi \langle\psi|\hat{H}^\lambda(v,j)|\psi\rangle = \inf_{(\sigma,\xi)} \left\{F^\lambda(\sigma,\xi) + v\sigma + j\xi\right\}$$

Universal "density" functional

$$\begin{aligned} F^\lambda(\sigma,\xi) &= \sup_{(v,j)} \left\{E^\lambda(v,j) - v\sigma - j\xi\right\} && \text{(Lieb)} \\ &= \inf_{\psi \mapsto (\sigma,\xi)} \langle\psi|\hat{H}_0^\lambda|\psi\rangle && \text{(pure-state Levy–Lieb)} \\ &= \inf_{\Gamma \mapsto (\sigma,\xi)} \operatorname{Tr} \hat{H}_0^\lambda \Gamma && \text{(mixed-state Levy–Lieb)} \end{aligned}$$

Properties of QRabi universal functional

(i) All functionals agree

(ii) The functional is convex and differentiable for all $(\sigma,\xi) \in (-1,1) \times \mathbb{R}$

(iii) Unique (pure-state) v-rep.: the Hohenberg–Kohn mapping

$$\mathbb{R} \times \mathbb{R} \ni (v,j) \longmapsto (\sigma,\xi) \in (-1,1) \times \mathbb{R}$$

is a *bijection*

(iv) Displacement rule

$$F^\lambda(\sigma,\xi) = F^\lambda(\sigma,0) + \lambda g\sigma\xi + \frac{\omega^2}{2}\xi^2$$

Hohenberg–Kohn Mapping in DFT

$$-v \in \underline{\partial} F^\lambda(\rho) = \left\{u \in X^* | \rho' \in X : F^\lambda(\rho') \geq F^\lambda(\rho) + \langle u, \rho' - \rho\rangle\right\}$$

$$-v \in \underline{\partial} F^\lambda(\rho) \iff \rho \in \overline{\partial} E^\lambda(v) \qquad (F^\lambda, E^\lambda) \text{ conjugate pair}$$

Hohenberg–Kohn Mapping for QRabi

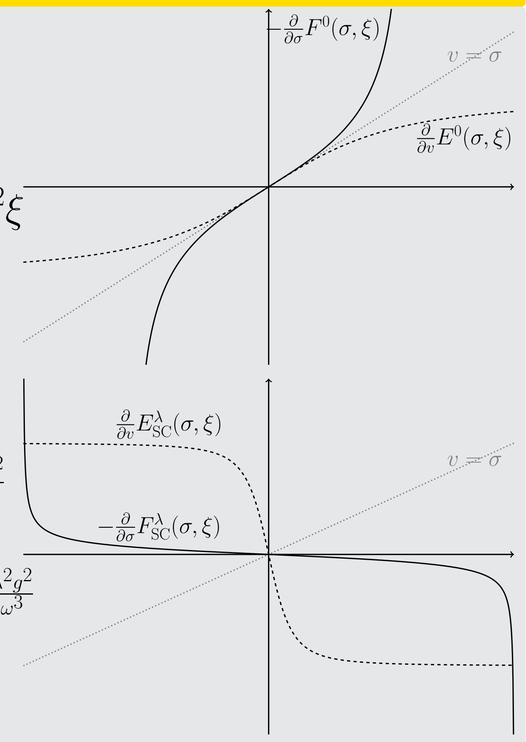
Zero coupling regime, $\lambda = 0$

$$F^0(\sigma,\xi) = \frac{\omega}{2} - t\sqrt{1-\sigma^2} + \frac{\omega^2}{2}\xi^2$$

$$v(\sigma) = -t\frac{\sigma}{\sqrt{1-\sigma^2}}, \quad \text{and} \quad j(\xi) = -\omega^2\xi$$

Strongly correlated regime, $t/\lambda \ll 1$

$$\begin{aligned} F_{\mathrm{SC}}^\lambda(\sigma,\xi) &\approx \frac{\omega}{2} + \frac{\omega^2}{2}\xi^2 + \lambda g\sigma\xi \\ &\quad - \lambda^2\frac{g^2}{2\omega^2}(1-\sigma^2) - t\sqrt{1-\sigma^2}e^{-\frac{\lambda^2 g^2}{\omega^3}} \\ v(\sigma,\xi) &\approx -\lambda g\xi - \frac{\lambda^2 g^2}{\omega^2}\sigma - t\frac{\sigma}{\sqrt{1-\sigma^2}}e^{-\frac{\lambda^2 g^2}{\omega^3}} \\ j(\sigma,\xi) &\approx -\omega^2\xi - \lambda g\sigma \end{aligned}$$



The Adiabatic Connection for QRabi

$\lambda \mapsto F^\lambda(\sigma,\xi)$ is concave

Newton–Leibniz formula

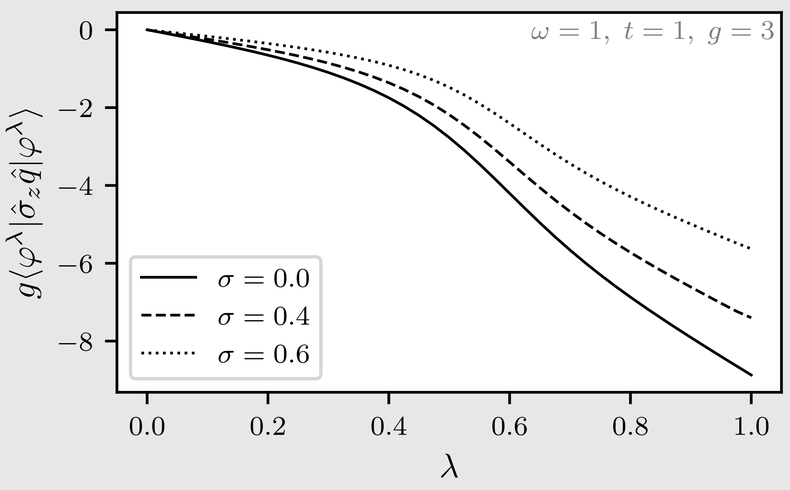
$$F^\lambda(\sigma,\xi) = F^0(\sigma,\xi) + \int_0^\lambda f^\tau(\sigma,\xi)\,\mathrm{d}\tau$$

where

$$\overline{\partial}_\lambda F^\lambda(\sigma,\xi) \ni f^\lambda(\sigma,\xi) = g\left\langle\psi^\lambda\middle|\hat{\sigma}_z\hat{q}\middle|\psi^\lambda\right\rangle$$

Adiabatic connection functional

$$\begin{aligned} F^\lambda(\sigma,\xi) = &\frac{\omega}{2} - t\sqrt{1-\sigma^2} + \frac{\omega^2}{2}\xi^2 \\ &+ \underbrace{\lambda g\sigma\xi}_{\lambda D(\sigma,\xi)} - \frac{\lambda^2 g^2}{2\omega^2}(1-\sigma^2) + I^\lambda(\sigma) \end{aligned}$$



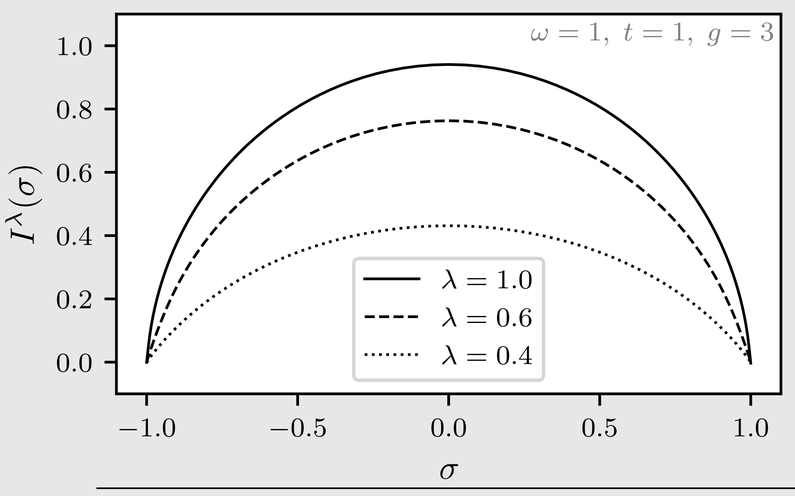
Non-explicit contribution

$$I^\lambda(\sigma) = -\frac{4tg}{\omega^2}\int_0^\lambda\int_{-\infty}^\infty \varphi_+^{\tau\prime}(q)\varphi_-^\tau(q)\,\mathrm{d}q\,\mathrm{d}\tau$$

$$\varphi^\lambda(q) = \begin{pmatrix}\varphi_+^\lambda(q)\\ \varphi_-^\tau(q)\end{pmatrix} \quad \text{optimiser of } F^\lambda(\sigma,0)$$

Conjecture

$$I^\lambda(\sigma) \approx \omega\frac{b}{a}\left(\sqrt{a^2-\sigma^2} - \sqrt{a^2-1}\right)$$



Exchange-correlation energy

$$\begin{aligned} \lambda E_{\mathrm{xc}}^\lambda(\sigma,\xi) &= \int_0^\lambda \left[f^\tau(\sigma,\xi) - D(\sigma,\xi)\right]\mathrm{d}\tau \\ &= \int_0^\lambda \left[g\left\langle\psi^\lambda\middle|\hat{\sigma}_z\hat{q}\middle|\psi^\lambda\right\rangle - g\sigma\xi\right]\mathrm{d}\tau \end{aligned}$$

Exchange energy

$$E_{\mathrm{x}}(\sigma,\xi) = \partial_\lambda^+ F^\lambda(\sigma,\xi)\big|_{\lambda=0} - D(\sigma,\xi) = 0$$

Correlation energy

$$\begin{aligned} \lambda E_{\mathrm{c}}(\sigma,\xi) &= \lambda E_{\mathrm{xc}}(\sigma,\xi) - \lambda E_{\mathrm{x}}(\sigma,\xi) \\ &= -\frac{\lambda^2 g^2}{2\omega^2}(1-\sigma^2) + I^\lambda(\sigma) \end{aligned}$$

Summary and outlook

- Instructive example
- Exact constraints
- Functional development
- (Inverse) Kohn–Sham and regularisation

References

Bakkestuen *et al.*, *J. Phys. Chem. A, 129, 9, 2337–2360* (2025)

Bakkestuen *et al.*, "Density-functional theory for the Dicke Hamiltonian" (2024) arXiv:2409.13767

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